

Logging using SLF4J

Exercise 1: Logging Error Messages and Warning Levels

Task: Write a Java application that demonstrates logging error messages and warning levels using SLF4J.

Code:

pom.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.sl4jdemo</groupId>
  <artifactId>SLF4JLoggingDemo</artifactId>
  <version>1.0-SNAPSHOT</version>

  <properties>
    <maven.compiler.source>17</maven.compiler.source>
    <maven.compiler.target>17</maven.compiler.target>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  </properties>
  <dependencies>

    <!-- SLF4J API -->
    <dependency>
      <groupId>org.slf4j</groupId>
      <artifactId>slf4j-api</artifactId>
      <version>1.7.30</version>
    </dependency>

    <!-- Logback Implementation -->
    <dependency>
      <groupId>ch.qos.logback</groupId>
      <artifactId>logback-classic</artifactId>
      <version>1.2.3</version>
    </dependency>

  </dependencies>

</project>
```

LoggingExample.java

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class LoggingExample {

    private static final Logger logger =
        LoggerFactory.getLogger(LoggingExample.class);

    public static void main(String[] args) {

        logger.error("This is an error message");

        logger.warn("This is a warning message");

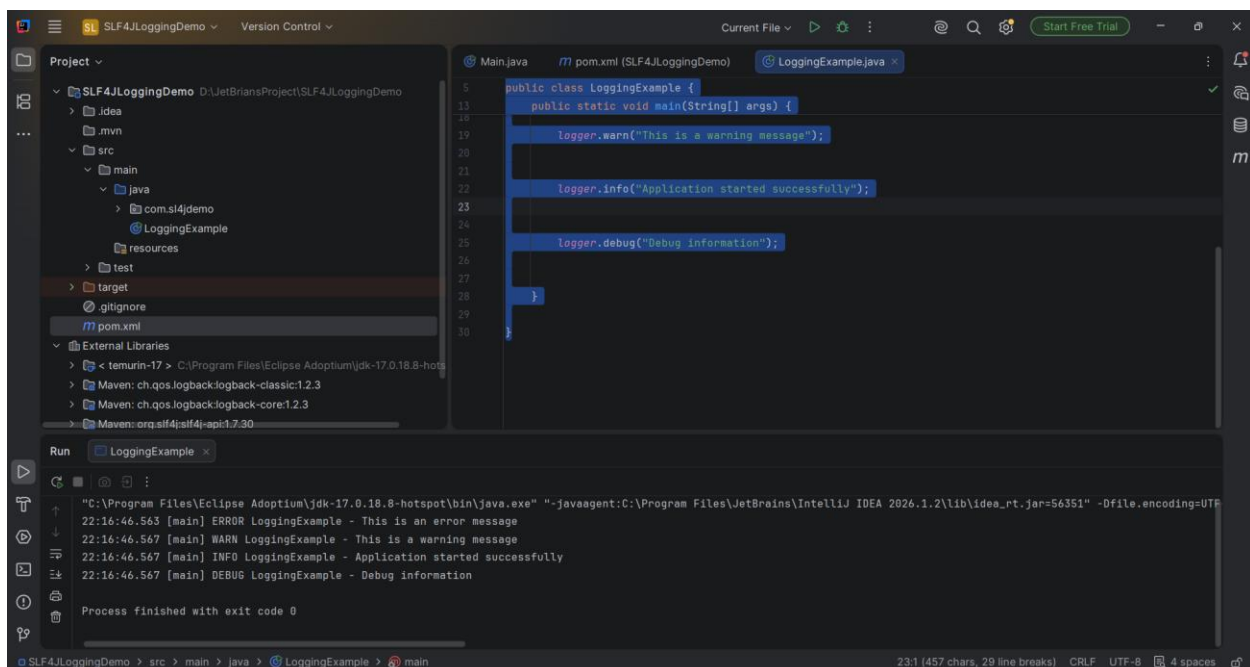
        logger.info("Application started successfully");

        logger.debug("Debug information");

    }

}
```

Output:



The screenshot shows an IDE window with the following components:

- Project View:** Shows the project structure for 'SLF4JLoggingDemo'. The 'src/main/java' directory contains 'com.sil4jdemo' which includes 'LoggingExample' and 'resources'.
- Code Editor:** Displays the 'LoggingExample.java' file. The code is as follows:

```
5 public class LoggingExample {
13     public static void main(String[] args) {
18
19         logger.warn("This is a warning message");
20
21
22         logger.info("Application started successfully");
23
24
25         logger.debug("Debug information");
26
27     }
28 }
29
30
```
- Run Console:** Shows the output of the program execution:

```
"C:\Program Files\Eclipse Adoptium\jdk-17.0.18-hotspot\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.2\lib\idea_rt.jar=56351" -Dfile.encoding=UTF-8
22:16:46.563 [main] ERROR LoggingExample - This is an error message
22:16:46.567 [main] WARN LoggingExample - This is a warning message
22:16:46.567 [main] INFO LoggingExample - Application started successfully
22:16:46.567 [main] DEBUG LoggingExample - Debug information
Process finished with exit code 0
```

Exercise 2: Parameterized Logging

Task: Write a Java application that demonstrates parameterized logging using SLF4J.

Code:

ParameterizedLoggingExample.java

```
package com.sl4jdemo;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class ParameterizedLoggingExample {

    private static final Logger logger =
        LoggerFactory.getLogger(ParameterizedLoggingExample.class);

    public static void main(String[] args) {

        String username = "Dave";
        int loginAttempts = 5;
        double accountBalance = 25000.75;

        // Parameterized logging using {}

        logger.info("User {} logged into the application", username);

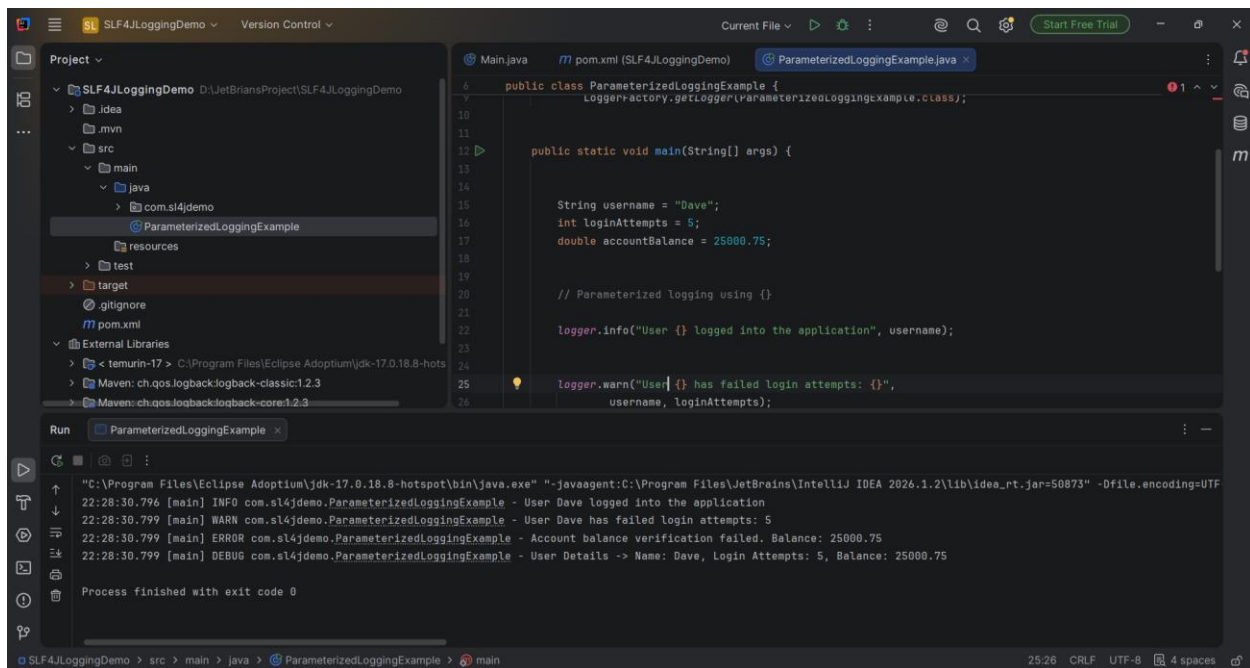
        logger.warn("User {} has failed login attempts: {}",
            username, loginAttempts);

        logger.error("Account balance verification failed. Balance: {}",
            accountBalance);

        logger.debug(
            "User Details -> Name: {}, Login Attempts: {}, Balance: {}",
            username,
            loginAttempts,
            accountBalance
        );

    }
}
```

Output:



Exercise 3: Using Different Appenders

Task: Write a Java application that demonstrates using different appenders with SLF4J.

Code:

logback.xml:

```

<?xml version="1.0" encoding="UTF-8"?>

<configuration>

    <!-- Console Appender -->
    <appender name="CONSOLE"
        class="ch.qos.logback.core.ConsoleAppender">

        <encoder>
            <pattern>
                %d{HH:mm:ss.SSS} [%thread] %-5level %logger{36} - %msg%n
            </pattern>
        </encoder>
    </appender>

    <!-- File Appender -->
    <appender name="FILE"
        class="ch.qos.logback.core.FileAppender">

```

```

        <file>app.log</file>

        <encoder>
            <pattern>
                %d{HH:mm:ss.SSS} [%thread] %-5level %logger{36} - %msg%n
            </pattern>
        </encoder>

    </appender>

    <!-- Root Logger -->
    <root level="debug">

        <appender-ref ref="CONSOLE"/>

        <appender-ref ref="FILE"/>

    </root>

</configuration>

```

AppenderExample.java

```

package com.sl4jdemo;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class AppenderExample {

    private static final Logger logger =
        LoggerFactory.getLogger(AppenderExample.class);

    public static void main(String[] args) {

        logger.debug("Application debugging started");

        logger.info("User logged in successfully");

        logger.warn("Memory usage is high");

        logger.error("Database connection failed");
    }
}

```

```
}  
  
}
```

Output:

The screenshot displays the IntelliJ IDEA IDE interface for a project named SLF4JLoggingDemo. The Project view on the left shows the project structure, including the source code, resources, and target directories. The main editor shows the AppenderExample.java file, which contains a simple logging application. The Run console at the bottom shows the output of the application, including the command used to run it and the resulting log messages.

```
Project  
├── com.sl4jdemo  
│   ├── AppenderExample  
│   ├── LoggingExample  
│   └── ParameterizedLoggingExample  
├── resources  
│   └── logback.xml  
├── test  
├── target  
├── .gitignore  
├── app.log  
├── pom.xml  
└── External Libraries  
    ├── temurin-17 > C:\Program Files\Eclipse Adoptium\jdk-17.0.18-hotspot  
    ├── Maven: ch.qos.logback:logback-classic:1.2.3  
    ├── Maven: ch.qos.logback:logback-core:1.2.3  
    └── Maven: org.slf4j:slf4j-api:1.7.30  
└── Scratches and Consoles
```

```
AppenderExample.java  
1 | 22:33:34.319 [main] DEBUG com.sl4jdemo.AppenderExample - Application debugging started  
2 | 22:33:34.320 [main] INFO com.sl4jdemo.AppenderExample - User logged in successfully  
3 | 22:33:34.321 [main] WARN com.sl4jdemo.AppenderExample - Memory usage is high  
4 | 22:33:34.321 [main] ERROR com.sl4jdemo.AppenderExample - Database connection failed  
5 |
```

```
Run  
"C:\Program Files\Eclipse Adoptium\jdk-17.0.18-hotspot\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.2\lib\idea_rt.jar=62273" -Dfile.encoding=UTF-8  
22:33:34.319 [main] DEBUG com.sl4jdemo.AppenderExample - Application debugging started  
22:33:34.320 [main] INFO com.sl4jdemo.AppenderExample - User logged in successfully  
22:33:34.321 [main] WARN com.sl4jdemo.AppenderExample - Memory usage is high  
22:33:34.321 [main] ERROR com.sl4jdemo.AppenderExample - Database connection failed  
Process finished with exit code 0
```